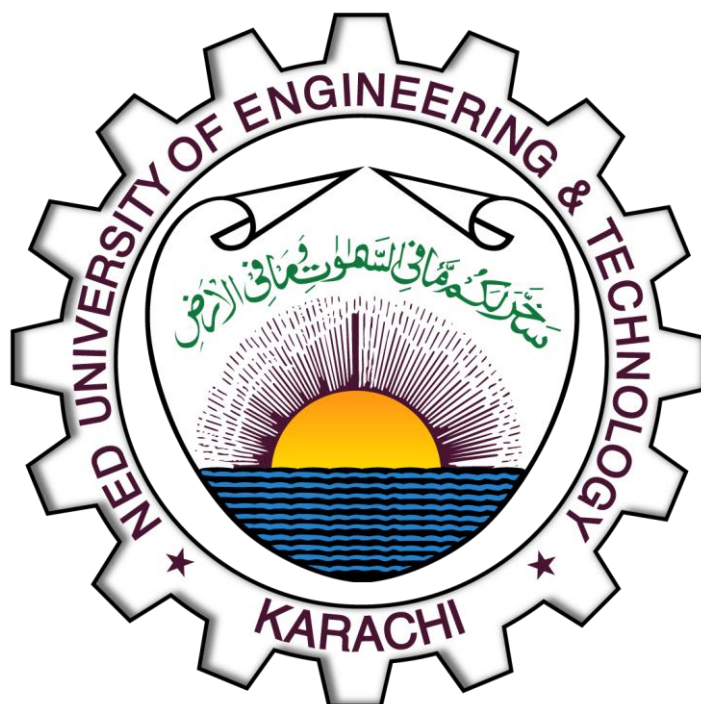


DEPARTMENT OF SOFTWARE

ENGINEERING

PROBABILITY AND STATISTICS

COURSE CODE:MT-331



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Central Limit Theorem (CLT)

The Central Limit Theorem (CLT) is a fundamental concept in probability and statistics. It states that when we take a sufficiently large number of independent and identically distributed (i.i.d.) random samples from any population with a finite mean and variance, the distribution of the sample mean becomes approximately normal, regardless of the original population distribution.

In simple words, even if the original data is not normally distributed, the average of many samples will tend to follow a normal distribution.

Suppose $X_1, X_2, X_3, \dots, X_n$ are independent and identically distributed random variables with:

- Mean = μ
- Variance = σ^2
- Then the sample mean is defined as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

The standardized form of the sample mean is:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

According to the Central Limit Theorem, as the sample size n becomes large, the distribution of Z approaches a standard normal distribution:

$$Z \sim N(0,1)$$

Conditions of CLT

For the Central Limit Theorem to hold, the following conditions should be satisfied:

- The random variables must be independent.
- They should be identically distributed.
- The population should have a finite mean and variance.
- The sample size should be sufficiently large (commonly $n \geq 30$).

Key Features

- The original population distribution can be normal or non-normal.
- As the sample size increases, the approximation to the normal distribution improves.
- The mean of the sampling distribution is equal to the population mean (μ).
- The standard deviation of the sampling distribution is $\frac{\sigma}{\sqrt{n}}$, also known as the standard error.

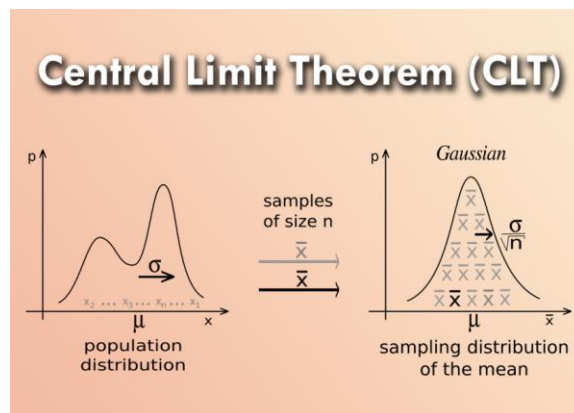
Importance and Applications

The Central Limit Theorem plays a crucial role in statistics and has wide applications:

- It forms the basis of hypothesis testing.
- It is used in constructing confidence intervals.
- It allows approximation of complex distributions using the normal distribution.
- It is widely applied in fields such as engineering, economics, data science, and quality control.

Conclusion

In conclusion, the Central Limit Theorem is a fundamental concept that simplifies statistical analysis by ensuring that the distribution of sample means approaches normality as the sample size increases. This makes it possible to apply powerful statistical tools even when the population distribution is unknown.



PROBLEM STATEMENT
API MONITORING SYSTEM

A large-scale web application records response times for millions of API requests daily. Due to varying network conditions and server load, individual response times fluctuate significantly and are not normally distributed.

From long-term monitoring, engineers have determined that:

- The average response time of individual requests is approximately **240 ms**
- The Variability in response time is about **80 ms**

To monitor performance, the system automatically takes random samples of **64 requests** every hour and computes the average response time for that hour.

Required:

- (a) Describe the distribution of the sample mean using CLT
- (b) Calculate the standard error of the sample mean
- (c) What is the probability that the hourly average response time exceeds **260 ms**?
- (d) Interpret the result in terms of system performance

Given

Population mean (μ) = 240 ms

Population standard deviation (σ) = 80 ms

Sample size (**n**) = 64

Solution

(a) Distribution of the Sample Mean

According to the **Central Limit Theorem (CLT)**, for large sample sizes, the sampling distribution of the mean is approximately normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting the given values:

$$\bar{X} \sim N\left(240, \frac{80}{\sqrt{64}}\right)$$

Since $\sqrt{64} = 8$:

$$\frac{80}{8} = 10$$

$$\therefore \bar{X} \sim N(240, 10)$$

Result: The sample mean follows a normal distribution with mean 240 ms and standard deviation 10 ms.

(b) Standard Error of the Mean

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{80}{\sqrt{64}} = \frac{80}{8} = 10$$

$$\boxed{SE = 10 \text{ ms}}$$

(c) Probability that Hourly Average Response Time Exceeds 260 ms

$$\bar{X} > 260$$

First, convert to standard normal variable:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{260 - 240}{10} = 2$$

Now,

$$P(\bar{X} > 260) = P(Z > 2)$$

$$P(Z > 2) = 1 - P(Z \leq 2)$$

From standard normal tables:

$$P(Z \leq 2) = 0.9772$$

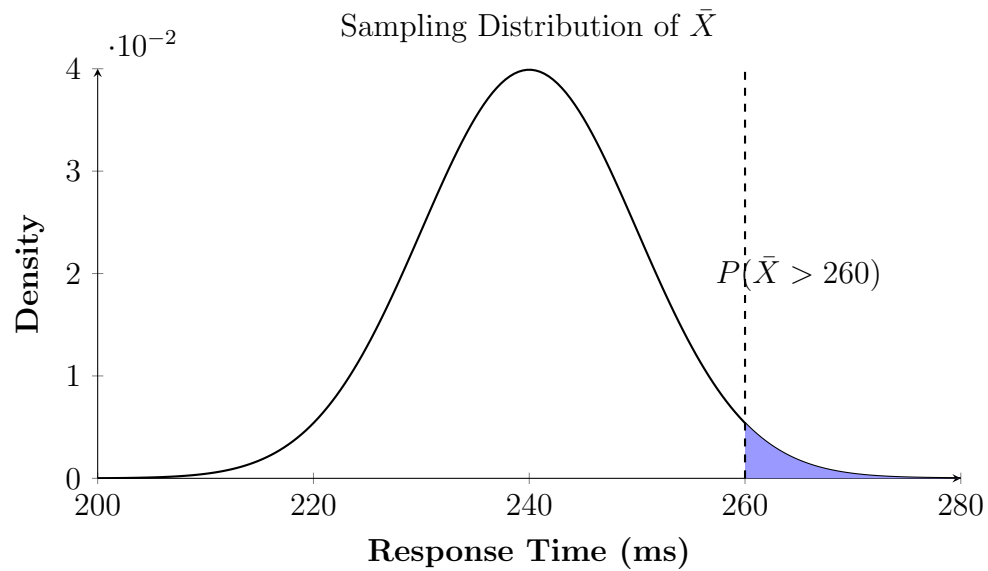
$$P(Z > 2) = 1 - 0.9772 = 0.0228$$

$$P(\bar{X} > 260) = 0.0228$$

(d) Interpretation

The probability of the average exceeding 260 ms is only 2.28%, which is very low. This means such an event is rare under normal conditions. Therefore, exceeding this threshold likely indicates abnormal system performance and can be used to trigger alerts.

Graphical Representation



PROBLEM STATEMENT

CONTINUOUS INTEGRATION (CI) PIPELINE ANALYSIS

In a software company, the time taken to complete a full CI pipeline (build + test + deploy) varies due to code complexity and server availability.

Historical data shows:

- Average pipeline completion time = **12 minutes**
- Standard deviation = **3 minutes**

A DevOps engineer randomly selects **36 pipeline executions** and computes the sample mean completion time.

Required:

- (a) What are the mean and standard deviation of the sampling distribution?
- (b) What is the probability that the average pipeline time is less than **11 minutes**?
- (c) What is the probability that the sample mean lies between **11.5** and **12.5 minutes**?
- (d) Explain how increasing the sample size would affect monitoring accuracy

Given

Population mean (μ) = 12 minutes

Population standard deviation (σ) = 3 minutes

Sample size (n) = 36

Solution

(a) Mean and Standard Deviation of Sampling Distribution

By the Central Limit Theorem (CLT), the sampling distribution of the mean is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting values:

$$\bar{X} \sim N\left(12, \frac{3}{\sqrt{36}}\right)$$

Since $\sqrt{36} = 6$:

$$\frac{3}{6} = 0.5$$

$$\therefore \bar{X} \sim N(12, 0.5)$$

Result:

- Mean = 12 minutes
- Standard deviation (Standard Error) = 0.5 minutes

(b) Probability that Average Pipeline Time is Less than 11 minutes

$$\bar{X} < 11$$

Standardizing:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{11 - 12}{0.5} = -2$$

Thus:

$$P(\bar{X} < 11) = P(Z < -2)$$

From Z-table:

$$P(Z < -2) = 0.0228$$

$$\boxed{P(\bar{X} < 11) = 0.0228}$$

(c) Probability that the Sample Mean lies between 11.5 and 12.5 minutes

$$11.5 \leq \bar{X} \leq 12.5$$

Convert both values to Z-scores:

$$Z_1 = \frac{11.5 - 12}{0.5} = -1 \quad , \quad Z_2 = \frac{12.5 - 12}{0.5} = 1$$

So:

$$P(11.5 \leq \bar{X} \leq 12.5) = P(-1 \leq Z \leq 1)$$

$$P(-1 \leq Z \leq 1) = P(Z < 1) - P(Z < -1)$$

From standard normal table:

$$P(Z < 1) = 0.8413, \quad P(Z < -1) = 0.1587$$

$$P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$$

$P(11.5 \leq \bar{X} \leq 12.5) = 0.6826$

(d) Effect of Increasing Sample Size

Increasing the sample size n reduces the standard error:

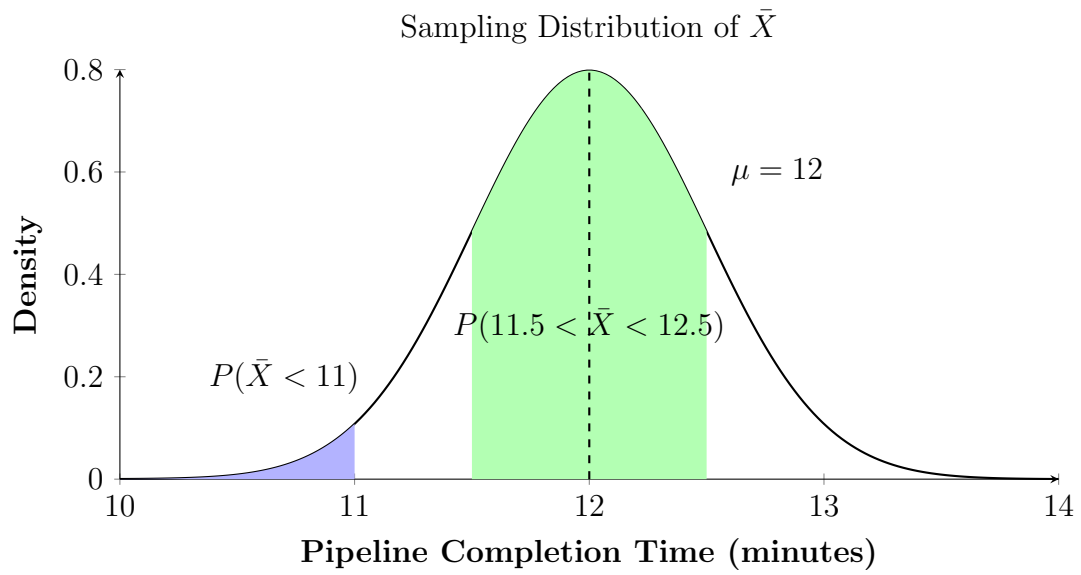
$$SE = \frac{\sigma}{\sqrt{n}}$$

As n increases:

- Standard error decreases
- Sampling distribution becomes narrower
- Sample mean becomes more stable and reliable
- Monitoring accuracy improves

Conclusion: Larger sample sizes lead to more precise estimates of CI pipeline performance and better detection of anomalies.

Graphical Representation



PROBLEM STATEMENT
BUG RESOLUTION IN AGILE SPRINT

In an agile development team, the time taken to resolve bugs varies depending on complexity, developer experience, and testing delays.

Analysis over several month shows:

- Mean resolution time = **18 hours**
- Standard deviation = **6 hours**

During a sprint review, a manager selects **49 randomly resolved bugs** and computes the sample mean resolution time.

Required:

- (a) Determine the standard error of the sample mean.
- (b) Find the probability that the average resolution time exceeds **20 hours**.
- (c) Find the probability that the average lies within ± 2 **hours** of the mean.
- (d) Comment on whether such variation is acceptable for sprint planning.

Given

Population mean (μ) = 180 hours

Population standard deviation (σ) = 6 hours

Sample size (**n**) = 49

Solution

(a) Standard Error of the Sample Mean

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{6}{\sqrt{49}}$$

Since $\sqrt{49} = 7$:

$$SE = \frac{6}{7} = 0.857$$

$$\boxed{SE \approx 0.86 \text{ hours}}$$

(b) Probability that Average Resolution Time Exceeds 20 Hours

Standardizing:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{20 - 18}{0.857} \approx 2.33$$

$$P(\bar{X} > 20) = P(Z > 2.33)$$

From Z-table:

$$P(Z < 2.33) = 0.9901$$

$$P(Z > 2.33) = 1 - 0.9901 = 0.0099$$

$$\boxed{P(\bar{X} > 20) \approx 0.0099}$$

(c) Probability that Average Lies Within ± 2 hours Of the Mean

Convert to Z-scores:

$$Z_1 = \frac{16 - 18}{0.857} \approx -2.33 \quad , \quad Z_2 = \frac{20 - 18}{0.857} \approx 2.33$$

$$P(16 < \bar{X} < 20) = P(-2.33 < Z < 2.33)$$

From Z-table:

$$P(Z < 2.33) = 0.9901, \quad P(Z < -2.33) = 0.0099$$

$$P(-2.33 < Z < 2.33) = 0.9901 - 0.0099 = 0.9802$$

$$\boxed{P(16 \leq \bar{X} \leq 20) \approx 0.9802}$$

(d) Interpretation for Sprint Planning

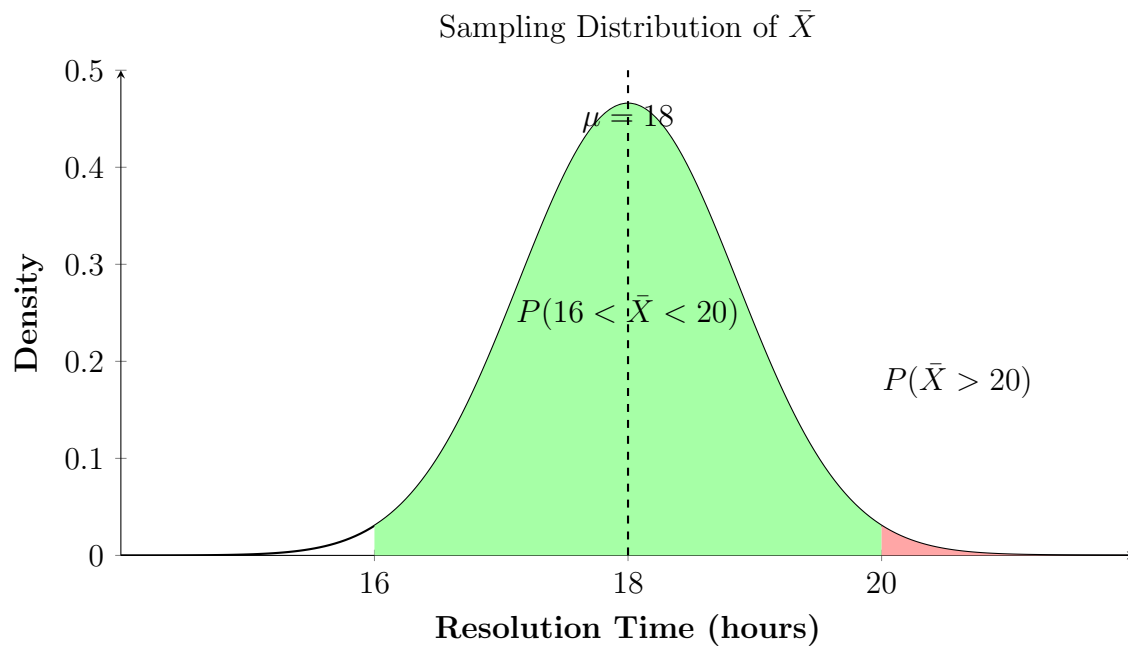
The probability that the sample mean stays within the range of 16 to 20 hours is approximately 98%, which is very high.

This indicates:

- Bug resolution times are relatively stable at the sprint level
- Small fluctuations are normal and expected
- Extreme delays (above 20 hours on average) are rare

Conclusion: The variation is acceptable for sprint planning, and the process is statistically stable.

Graphical Representation



PROBLEM STATEMENT
DATABASE QUERY PERFORMANCE

A database system processes a large number of queries. The execution times of individual queries are right-skewed due to occasional heavy queries. However, historical data shows:

- Population mean execution time: **4 seconds**
- Population standard deviation: **1.5 seconds**
- Sample size used for analysis: **100 queries**

A performance engineer takes random samples of **100 queries** to analyze system performance.

Required:

- (a) Using the Central Limit Theorem, state the mean and standard deviation of the sampling distribution of the sample mean execution time.
- (b) Calculate the standard error of the sample mean execution time.
- (c) Find the probability that the sample mean execution time is greater than **4.3 seconds**.
- (d) Find the probability that the sample mean execution time lies between **3.8 seconds** and **4.2 seconds**.
- (e) Explain why the Central Limit Theorem is useful in analyzing this system even though the original data is skewed.

Given

Population mean (μ) = 4 seconds

Population standard deviation (σ) = 1.5 seconds

Sample size (**n**) = 100

Solution

(a) Distribution of the Sample Mean (CLT)

According to the **Central Limit Theorem (CLT)**, for large sample sizes, the sampling distribution of the mean is approximately normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting values:

$$\bar{X} \sim N\left(4, \frac{1.5}{\sqrt{100}}\right)$$

Since $\sqrt{100} = 10$:

$$\frac{1.5}{10} = 0.15$$

$$\therefore \bar{X} \sim N(4, 0.15)$$

Result: The sample mean follows a normal distribution with mean **4 seconds** and standard deviation **0.15 seconds**.

(b) Standard Error of the Sample Mean

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{1.5}{\sqrt{100}} = \frac{1.5}{10} = 0.15$$

$$\boxed{SE = 0.15 \text{ seconds}}$$

(c) Probability that Sample Mean Exceeds 4.3 seconds

$$\bar{X} > 4.3$$

Convert to standard normal variable:

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{4.3 - 4}{0.15} = \frac{0.3}{0.15} = 2$$

$$P(\bar{X} > 4.3) = P(Z > 2) = 1 - P(Z \leq 2)$$

From standard normal tables:

$$P(Z \leq 2) = 0.9772$$

$$P(Z > 2) = 1 - 0.9772 = 0.0228$$

$$\boxed{P(\bar{X} > 4.3) = 0.0228}$$

(d) Probability that Sample Mean Lies Between 3.8 and 4.2 seconds

$$3.8 \leq \bar{X} \leq 4.2$$

Convert to Z-scores:

$$Z_1 = \frac{3.8 - 4}{0.15} = \frac{-0.2}{0.15} \approx -1.33 \quad , \quad Z_2 = \frac{4.2 - 4}{0.15} = \frac{0.2}{0.15} \approx 1.33$$

$$\begin{aligned} P(3.8 \leq \bar{X} \leq 4.2) &= P(-1.33 \leq Z \leq 1.33) \\ &= P(Z < 1.33) - P(Z < -1.33) \end{aligned}$$

From standard normal tables:

$$P(Z < 1.33) = 0.9082, \quad P(Z < -1.33) = 0.0918$$

$$P(-1.33 < Z < 1.33) = 0.9082 - 0.0918 = 0.8164$$

$$\boxed{P(3.8 \leq \bar{X} \leq 4.2) = 0.8164}$$

(e) Why CLT is Useful Despite Skewed Data

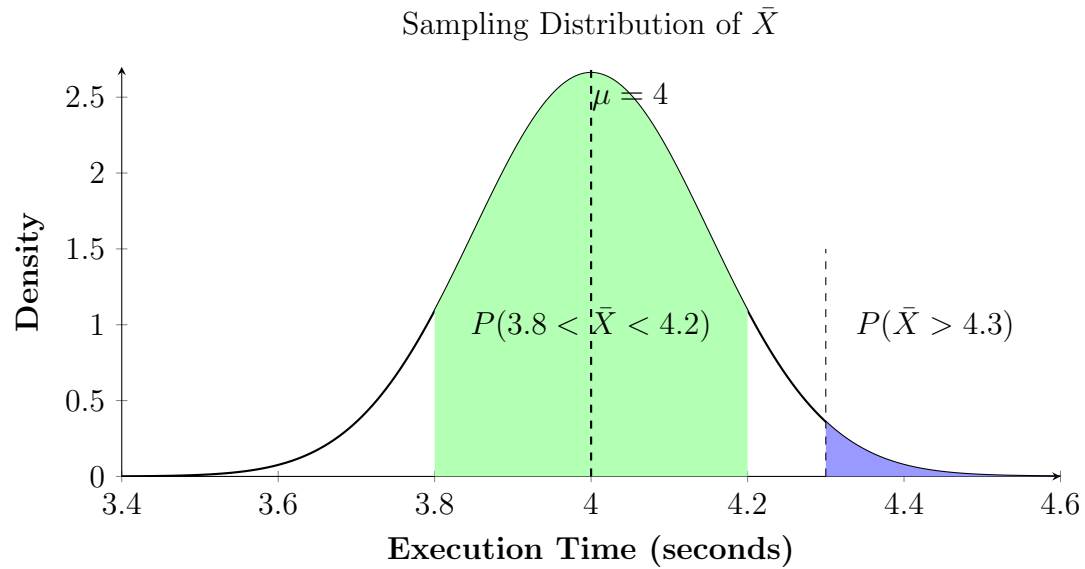
Although individual query execution times are right-skewed (due to occasional heavy queries), the **Central Limit Theorem** guarantees that the sampling distribution of the sample mean approaches a **normal distribution** as the sample size increases, regardless of the original population's shape.

With $n = 100$ (a sufficiently large sample):

- The distribution of $\bar{X}$ is approximately normal
- Standard probability methods (Z-scores, normal tables) can be applied
- Performance thresholds can be evaluated using well-understood normal distribution properties

Conclusion: The CLT allows engineers to make reliable statistical inferences about average query performance without requiring the raw data to be normally distributed.

Graphical Representation



PROBLEM STATEMENT

CLOUD SYSTEM LOAD ANALYSIS

A cloud platform tracks the number of active users per minute. The number fluctuates due to user behavior and time of day. Statistics show:

- Mean number of users = **1500**
- Standard deviation = **300**

To estimate system load, engineers compute the average number of users over **25 randomly selected minutes**.

Required:

- (a) Calculate the standard error.
- (b) Find the probability that the average number of users exceeds **1600**.
- (c) Find the probability that the average lies between **1400** and **1550** users.
- (d) Explain how this helps in capacity planning.

Given

Population mean (μ) = 1500 users

Population standard deviation (σ) = 300 users

Sample size (**n**) = 25

Solution

(a) Standard Error

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{300}{\sqrt{25}} = \frac{300}{5} = 60$$

$$\boxed{SE = 60 \text{ users}}$$

By the CLT, the sampling distribution is:

$$\bar{X} \sim N(1500, 60)$$

(b) Probability that Average Number of Users Exceeds 1600

$$\bar{X} > 1600$$

Convert to Z-score:

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{1600 - 1500}{60} = \frac{100}{60} \approx 1.67$$

$$P(\bar{X} > 1600) = P(Z > 1.67) = 1 - P(Z \leq 1.67)$$

From standard normal tables:

$$P(Z \leq 1.67) = 0.9525$$

$$P(Z > 1.67) = 1 - 0.9525 = 0.0475$$

$$\boxed{P(\bar{X} > 1600) = 0.0475}$$

(c) Probability that Average Lies Between 1400 and 1550 Users

$$1400 \leq \bar{X} \leq 1550$$

Convert to Z-scores:

$$Z_1 = \frac{1400 - 1500}{60} = \frac{-100}{60} \approx -1.67 \quad , \quad Z_2 = \frac{1550 - 1500}{60} = \frac{50}{60} \approx 0.83$$

$$\begin{aligned} P(1400 \leq \bar{X} \leq 1550) &= P(-1.67 \leq Z \leq 0.83) \\ &= P(Z < 0.83) - P(Z < -1.67) \end{aligned}$$

From standard normal tables:

$$P(Z < 0.83) = 0.7967, \quad P(Z < -1.67) = 0.0475$$

$$P(-1.67 < Z < 0.83) = 0.7967 - 0.0475 = 0.7492$$

$$\boxed{P(1400 \leq \bar{X} \leq 1550) = 0.7492}$$

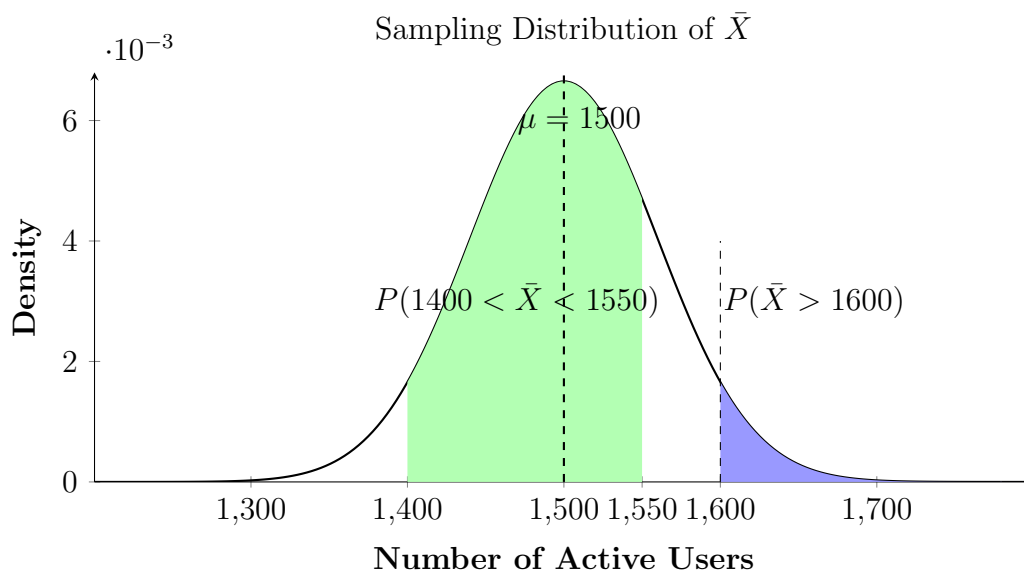
(d) How This Helps in Capacity Planning

By computing the sampling distribution of average active users, engineers can:

- Estimate the range within which the average load typically falls
- Identify the probability of load spikes exceeding critical thresholds (e.g., 1600 users)
- Provision servers and resources accordingly to handle expected peak demand
- Set automated alert thresholds at statistically rare values to detect anomalies

Conclusion: Statistical analysis of sample means allows the cloud platform team to proactively manage infrastructure capacity and minimize downtime risk.

Graphical Representation



PROBLEM STATEMENT

CODE REVIEW SYSTEM PERFORMANCE ANALYSIS

A software engineering team is analyzing the efficiency of its code review process. The time taken to complete a code review varies across pull requests due to differences in complexity and size of changes. The distribution of individual review times is **not normal**.

From historical data:

- Population mean code review time (μ) = **10 hours**
- Population standard deviation (σ) = **4 hours**

A random sample of $n = 49$ pull requests is selected, and the sample mean review time is analyzed.

Required:

- Find the mean and standard error of the sampling distribution of the sample mean code review time.
- What is the probability that the sample mean code review time lies between **9 hours** and **11 hours**?
- What is the probability that the sample mean code review time exceeds **11 hours**?
- Explain how the Central Limit Theorem (CLT) applies in this situation and why it allows the use of a normal distribution for the sample mean in analyzing code review performance in software engineering.

Given

Population mean (μ) = 10 hours

Population standard deviation (σ) = 4 hours

Sample size (n) = 49

Solution

(a) Mean and Standard Error of the Sampling Distribution

By the **Central Limit Theorem (CLT)**, the sampling distribution of the mean is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting values:

$$\bar{X} \sim N\left(10, \frac{4}{\sqrt{49}}\right)$$

Since $\sqrt{49} = 7$:

$$SE = \frac{4}{7} \approx 0.571$$

$$\therefore \bar{X} \sim N(10, 0.571)$$

Result:

- Mean of sampling distribution = **10 hours**
- Standard Error (SE) $\approx$ **0.571 hours**

$$\boxed{SE \approx 0.57 \text{ hours}}$$

(b) Probability that Sample Mean Lies Between 9 and 11 hours

$$9 \leq \bar{X} \leq 11$$

Convert to Z-scores:

$$Z_1 = \frac{9 - 10}{0.571} \approx -1.75 \quad , \quad Z_2 = \frac{11 - 10}{0.571} \approx 1.75$$

$$\begin{aligned} P(9 \leq \bar{X} \leq 11) &= P(-1.75 \leq Z \leq 1.75) \\ &= P(Z < 1.75) - P(Z < -1.75) \end{aligned}$$

From standard normal tables:

$$P(Z < 1.75) = 0.9599, \quad P(Z < -1.75) = 0.0401$$

$$P(-1.75 < Z < 1.75) = 0.9599 - 0.0401 = 0.9198$$

$$\boxed{P(9 \leq \bar{X} \leq 11) \approx 0.9198}$$

(c) **Probability that Sample Mean Exceeds 11 hours**

$$\bar{X} > 11$$

Convert to Z-score:

$$Z = \frac{11 - 10}{0.571} \approx 1.75$$

$$P(\bar{X} > 11) = P(Z > 1.75) = 1 - P(Z \leq 1.75)$$

From standard normal tables:

$$P(Z \leq 1.75) = 0.9599$$

$$P(Z > 1.75) = 1 - 0.9599 = 0.0401$$

$P(\bar{X} > 11) \approx 0.0401$

(d) **Application of CLT to Code Review Performance Analysis**

Even though individual code review times are **not normally distributed** (they may be skewed due to complex pull requests or developer workload spikes), the **Central Limit Theorem** guarantees:

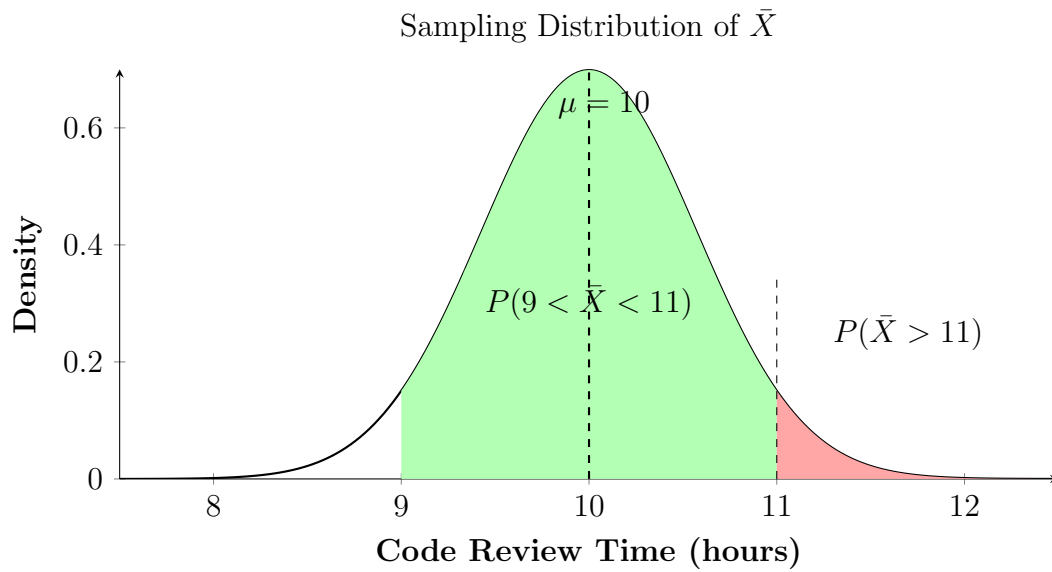
- With a sufficiently large sample size ($n = 49 \geq 30$), the sampling distribution of the sample mean $\bar{X}$ is **approximately normal**, regardless of the shape of the original distribution.
- The mean of the sampling distribution equals the population mean: $\mu_{\bar{X}} = \mu = 10$ hours.
- The spread of the sampling distribution is reduced by a factor of $\sqrt{n}$: $SE = \sigma/\sqrt{n} \approx 0.57$ hours.

In software engineering :

- Teams can use normal probability calculations to estimate the likelihood of average review times falling within acceptable limits.
- Delays exceeding 11 hours on average occur with only $\approx 4\%$ probability, meaning they are rare under normal conditions.
- If the observed sample mean significantly exceeds expected values, it signals a bottleneck in the review process requiring managerial action.

Conclusion: The CLT is essential in software process analysis it transforms a skewed real-world distribution into a tractable normal model, enabling data-driven decisions in sprint planning, resource allocation, and process improvement.

Graphical Representation



PROBLEM STATEMENT

MICROSERVICES MEMORY LEAK DETECTION

In a cloud-native application, hundreds of microservices run simultaneously across Kubernetes pods. Due to varying workloads, garbage collection cycles, and memory allocation patterns, the memory usage of individual pods fluctuates significantly and is not normally distributed.

From infrastructure monitoring over several months, the DevOps team has determined that:

- The average memory consumption per pod is approximately **512 MB**
- The variability in memory usage is about **96 MB** (standard deviation)

To detect potential memory leaks early, the system automatically samples **64 pods** every 30 minutes and computes the average memory usage across that batch.

Required:

1. Describe the distribution of the sample mean using CLT
2. Calculate the standard error of the sample mean
3. What is the probability that the 30-minute batch average memory usage exceeds **524 MB**?
4. Interpret the result in terms of memory leak alerting and pod auto-scaling decisions

Given

$$\begin{aligned}\text{Population mean } (\mu) &= 512 \text{ MB} \\ \text{Population standard deviation } (\sigma) &= 96 \text{ MB} \\ \text{Sample size } (n) &= 64\end{aligned}$$

Solution

(a) Distribution of the Sample Mean

According to the **Central Limit Theorem (CLT)**, for large sample sizes, the sampling distribution of the mean is approximately normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting the given values:

$$\bar{X} \sim N\left(512, \frac{96}{\sqrt{64}}\right)$$

Since $\sqrt{64} = 8$:

$$\frac{96}{8} = 12$$

$$\therefore \bar{X} \sim N(512, 12)$$

Result: The sample mean follows a normal distribution with mean 512 MB and standard deviation 12 MB, regardless of the underlying non-normal distribution of individual pods, because $n = 64 \geq 30$.

(b) Standard Error of the Mean

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{96}{\sqrt{64}} = \frac{96}{8} = 12$$

$$\boxed{SE = 12 \text{ MB}}$$

(c) Probability that Batch Average Memory Exceeds 524 MB

$$\bar{X} > 524$$

First, convert to standard normal variable:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{524 - 512}{12} = \frac{12}{12} = 1$$

Now,

$$P(\bar{X} > 524) = P(Z > 1)$$

$$P(Z > 1) = 1 - P(Z \leq 1)$$

From standard normal tables:

$$P(Z \leq 1) = 0.8413$$

$$P(Z > 1) = 1 - 0.8413 = 0.1587$$

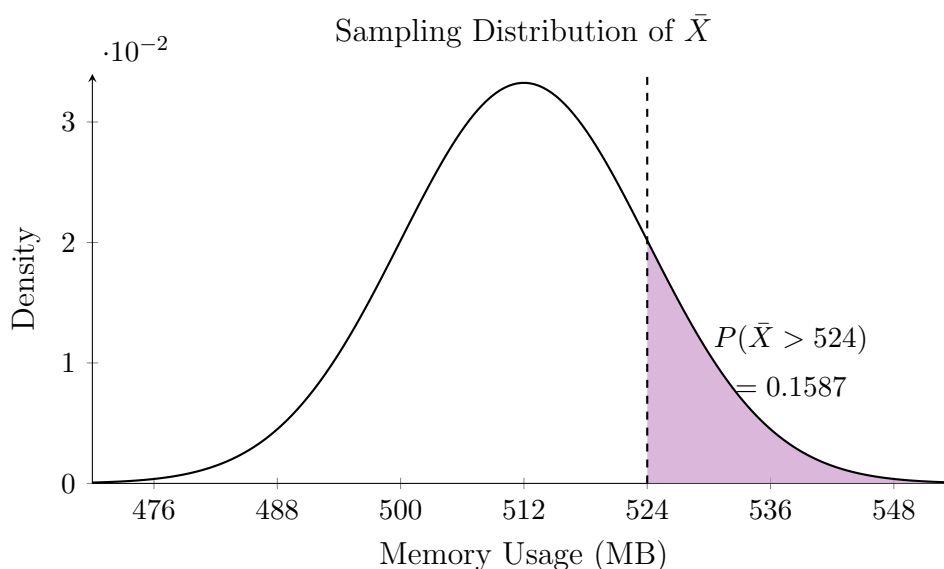
$$\boxed{P(\bar{X} > 524) = 0.1587}$$

(d) Interpretation

The probability that the 30-minute batch average memory usage exceeds 524 MB is **15.87%**, which is moderately high. This means such spikes can occur fairly regularly under normal load patterns and do not alone confirm a memory leak.

Setting a hard alert at 524 MB would generate false positives roughly 1 in 6 monitoring windows, causing alert fatigue. A better threshold is $\mu + 2SE = 512 + 24 = 536$ MB (corresponding to $Z = 2$, probability = 2.28%), making alerts genuinely exceptional. When the batch mean consistently exceeds 524 MB across multiple windows, the Kubernetes Horizontal Pod Autoscaler (HPA) can be triggered to redistribute memory load across additional pod replicas.

Graphical Representation



PROBLEM STATEMENT

REAL-TIME EVENT STREAM PROCESSING LATENCY

A fintech company operates a high-throughput event streaming pipeline (built on Apache Kafka) that processes millions of financial transactions per day. Due to varying message sizes, broker rebalancing events, and consumer group lag, the end-to-end processing latency of individual events is **highly right-skewed** and far from normally distributed.

Long-term telemetry data from the pipeline reveals that:

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- The average end-to-end latency per event is **35 ms**
- The standard deviation of latency is **14 ms**

To ensure SLA compliance, the SRE team randomly samples **49 events** every 15 minutes and computes the average latency for that window.

Required:

1. Using CLT, describe the sampling distribution of the average latency across the 49-event window
2. Calculate the standard error of the sample mean
3. What is the probability that the windowed average latency exceeds **39 ms**? (SLA breach threshold)
4. Explain how this statistical approach helps the SRE team make data-driven SLA breach alerts rather than reacting to individual noisy spikes

Given

Population mean (μ) = 35 ms

Population standard deviation (σ) = 14 ms

Sample size (n) = 49

Solution

(a) Distribution of the Sample Mean

According to the **Central Limit Theorem (CLT)**, even though individual event latencies follow a right-skewed distribution, the sampling distribution of the mean for $n = 49$ (≥ 30) is approximately normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting the given values:

$$\bar{X} \sim N\left(35, \frac{14}{\sqrt{49}}\right)$$

Since $\sqrt{49} = 7$:

$$\frac{14}{7} = 2$$

$$\therefore \bar{X} \sim N(35, 2)$$

Result: The 15-minute windowed average latency follows a normal distribution with mean **35 ms** and standard deviation **2 ms**. The CLT justifies this approximation despite the highly skewed individual latency distribution.

(b) Standard Error of the Mean

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{14}{\sqrt{49}} = \frac{14}{7} = 2$$

$$\boxed{SE = 2 \text{ ms}}$$

(c) Probability that Windowed Average Latency Exceeds 39 ms

$$\bar{X} > 39$$

First, convert to standard normal variable:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{39 - 35}{2} = \frac{4}{2} = 2$$

Now,

$$P(\bar{X} > 39) = P(Z > 2)$$

$$P(Z > 2) = 1 - P(Z \leq 2)$$

From standard normal tables:

$$P(Z \leq 2) = 0.9772$$

$$P(Z > 2) = 1 - 0.9772 = 0.0228$$

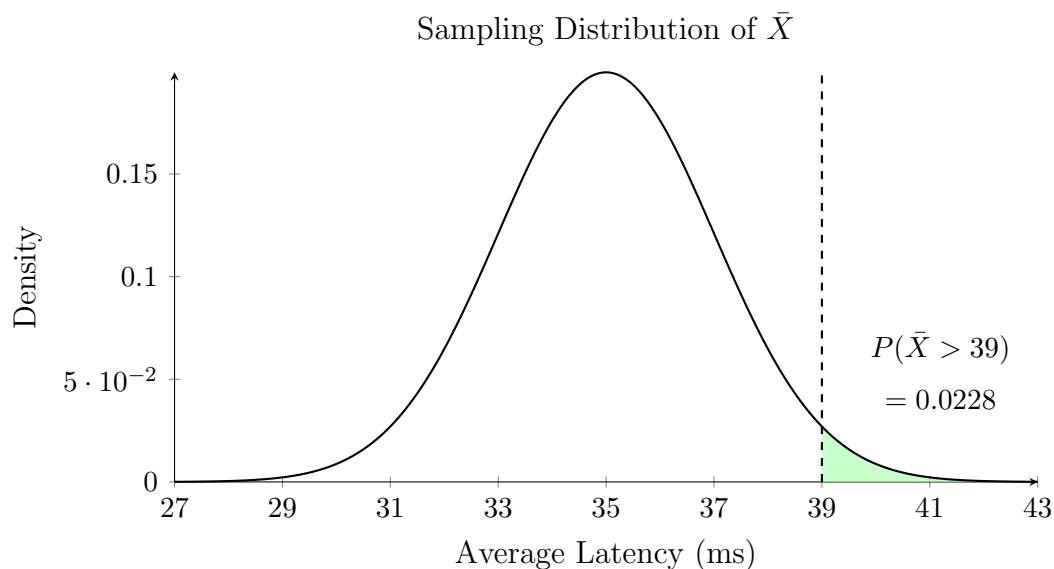
$$\boxed{P(\bar{X} > 39) = 0.0228}$$

(d) Interpretation

The probability that the windowed average latency exceeds the SLA threshold of 39 ms is only **2.28%**. This is very low under normal pipeline conditions, making such an event a genuine statistical signal rather than random noise.

Individual Kafka events may spike to 100 ms or more due to a temporary broker rebalance or garbage collection pause — reacting to each individual spike would produce constant false alerts. By averaging 49 events per window, the SRE team leverages the CLT to smooth out variance: $SE = 2$ ms is dramatically smaller than $\sigma = 14$ ms. With $P(\bar{X} > 39) = 2.28\%$, alerts at the 39 ms threshold fire approximately once in every 44 monitoring windows under healthy conditions — a strong, actionable signal of genuine degradation rather than transient noise.

Graphical Representation



PROBLEM STATEMENT
IoT SMART SENSOR DEPLOYMENT

A tech startup has deployed thousands of smart temperature sensors across a metropolitan area. Due to varying exposure to sunlight and wind, the battery life of individual sensors fluctuates significantly and does not follow a normal distribution.

From long-term field data, engineers have determined that:

- The average battery life of individual sensors is approximately **800 days**
- The variability in battery life is about **120 days**

To monitor the health of the network, a maintenance engineer randomly selects a batch of **64 sensors** and calculates their average remaining life.

Required:

- (a) Describe the distribution of the sample mean using CLT
- (b) Calculate the standard error of the sample mean
- (c) What is the probability that the average battery life for this sample is less than **770 days**?
- (d) Interpret the result in terms of system performance alerts

Given

Population mean (μ) = 800 days

Population standard deviation (σ) = 120 days

Sample size (**n**) = 64

Solution

(a) Distribution of the Sample Mean

According to the **Central Limit Theorem (CLT)**, for large sample sizes, the sampling distribution of the mean is approximately normal:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting the given values:

$$\bar{X} \sim N\left(800, \frac{120}{\sqrt{64}}\right)$$

Since $\sqrt{64} = 8$:

$$\frac{120}{8} = 15$$

$$\therefore \bar{X} \sim N(800, 15)$$

Result: Even though individual sensor data is not normally distributed, the sample mean of 64 sensors follows a normal distribution with mean 800 days and standard deviation 15 days.

(b) Standard Error of the Mean

The standard error is given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{120}{\sqrt{64}} = \frac{120}{8} = 15$$

$$\boxed{SE = 15 \text{ days}}$$

Meaning: The sample mean typically fluctuates by ± 15 days around the true mean, showing moderate stability.

(c) Probability that Sample Mean < 770 days

$$\bar{X} < 770$$

First, convert to standard normal variable:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{770 - 800}{15} = \frac{-30}{15} = -2$$

Now,

$$P(\bar{X} < 770) = P(Z < -2)$$

Directly from the standard normal table (left-tail cumulative):

$$P(Z < -2) = 0.0228$$

$$P(\bar{X} < 770) = 0.0228$$

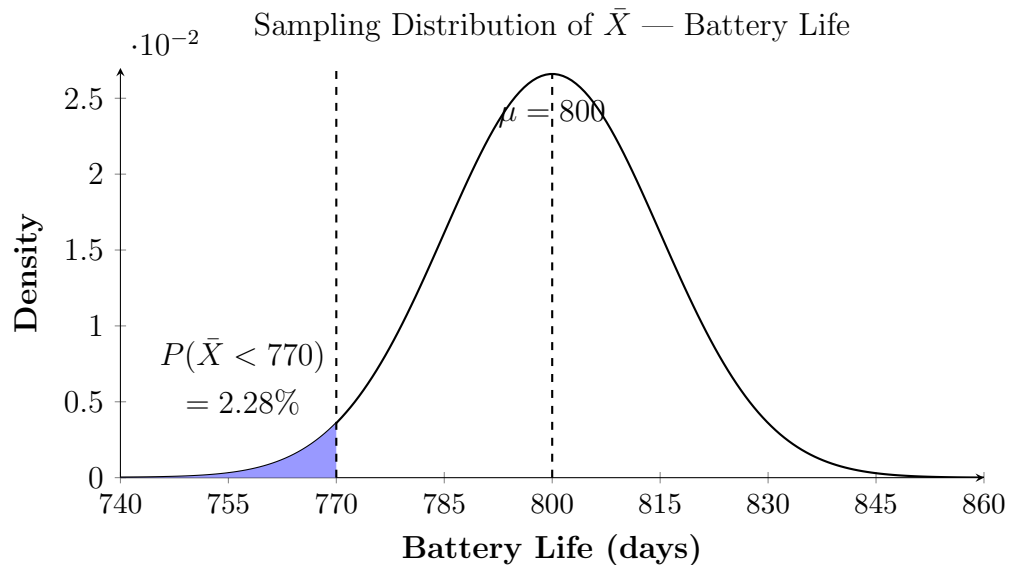
(d) Interpretation: System Performance Alerts

The probability of the average battery life dropping below 770 days is only 2.28%, which is very low. This means such an event is rare under normal operating conditions. Therefore:

- If such a drop occurs, it likely indicates abnormal system behavior
- Possible causes include battery degradation, environmental stress, or hardware faults
- This threshold can be used to trigger automatic maintenance alerts in the monitoring system

Conclusion: This is a very low probability, meaning a drop in average battery life below 770 days is unusual and should prompt immediate system investigation.

Graphical Representation



Graph Interpretation: The shaded left tail represents the small probability area where average battery life falls below 770 days. This rare event confirms that such a drop is highly unusual under normal conditions and warrants a system alert.

PROBLEM STATEMENT
SERVER RACK POWER CONSUMPTION

In a large data center, the power consumption (in Watts) of individual servers varies based on the computational load and the efficiency of the cooling fans. The distribution of power usage is non-normal due to some servers being idle while others are running high-intensity AI training models.

Historical records show that:

- The average power consumption per server is approximately **400 W**
- The variability in power consumption is about **50 W**

A facility manager observes a cluster of **100 servers** and computes the average power usage for the cluster.

Required:

- (a) What are the mean and standard deviation (standard error) of the sampling distribution?
- (b) What is the probability that the average power consumption is less than **392 W**?
- (c) What is the probability that the sample mean lies between **395 W** and **405 W**?
- (d) Explain how increasing the cluster size (sample size) would affect monitoring accuracy

Given

Population mean (μ) = 400 W

Population standard deviation (σ) = 50 W

Sample size (**n**) = 100

Solution

(a) Mean and Standard Deviation of the Sampling Distribution

By the Central Limit Theorem (CLT), the sampling distribution of the mean is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Substituting values:

$$\bar{X} \sim N\left(400, \frac{50}{\sqrt{100}}\right)$$

Since $\sqrt{100} = 10$:

$$\frac{50}{10} = 5$$

$$\therefore \bar{X} \sim N(400, 5)$$

Result:

- Mean ($\mu_{\bar{X}}$) = 400 W
- Standard Error (SE) = 5 W

Conclusion: The cluster's average power usage is highly stable with only 5 W variation.

(b) Probability that Average Power Consumption < 392 W

$$\bar{X} < 392$$

Standardizing:

$$Z = \frac{\bar{X} - \mu}{SE}$$

$$Z = \frac{392 - 400}{5} = \frac{-8}{5} = -1.6$$

Thus:

$$P(\bar{X} < 392) = P(Z < -1.6)$$

Directly from the standard normal table (left-tail cumulative):

$$P(Z < -1.6) = 0.0548$$

$$\boxed{P(\bar{X} < 392) = 0.0548}$$

Conclusion: There is a low chance of such a drop, indicating unusual low power usage in the cluster.

(c) Probability that Sample Mean lies between 395 W and 405 W

$$395 \leq \bar{X} \leq 405$$

Convert both values to Z-scores:

$$Z_1 = \frac{395 - 400}{5} = -1 \quad , \quad Z_2 = \frac{405 - 400}{5} = 1$$

So:

$$P(395 \leq \bar{X} \leq 405) = P(-1 \leq Z \leq 1)$$

$$P(-1 \leq Z \leq 1) = P(Z < 1) - P(Z < -1)$$

From standard normal table:

$$P(Z < 1) = 0.8413, \quad P(Z < -1) = 0.1587$$

$$P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$$

$$\boxed{P(395 < \bar{X} < 405) = 0.6826}$$

Conclusion: About 68.26% of the time, average power consumption stays within this range, showing stable and predictable system behavior.

(d) Effect of Increasing Sample Size

Increasing the sample size n reduces the standard error:

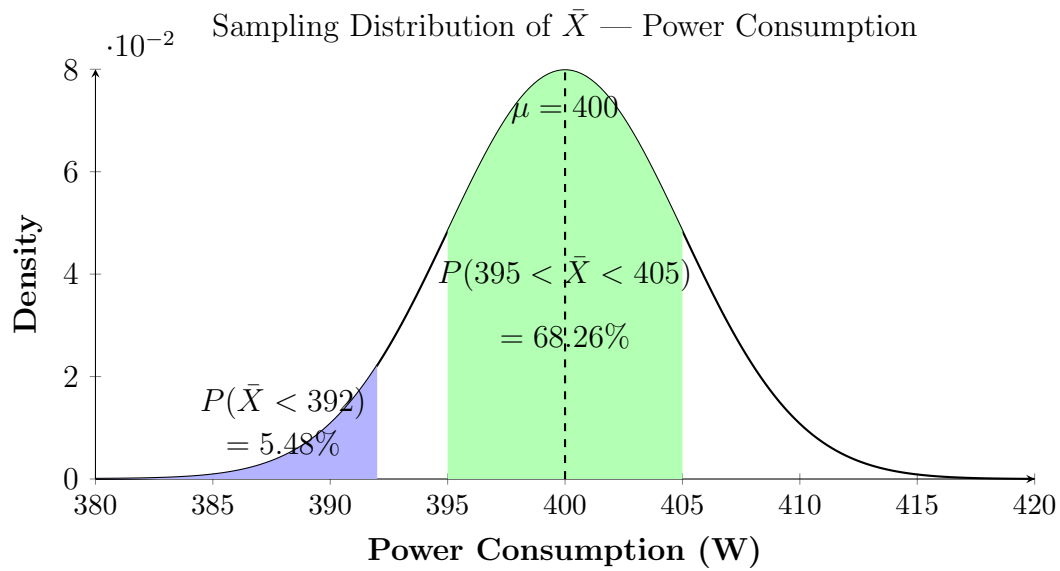
$$SE = \frac{\sigma}{\sqrt{n}}$$

As n increases:

- Denominator ($\sqrt{n}$) increases
- Standard error decreases
- Sampling distribution becomes narrower and tighter
- Sample mean becomes more stable and reliable
- Monitoring accuracy improves significantly

Conclusion: Larger sample sizes lead to more precise estimates of average power consumption and better detection of anomalies in data centers.

Graphical Representation



Graph Interpretation: The green shaded central region shows that most values lie near the mean, indicating stable and predictable system performance. The blue left tail represents the rare case of unusually low power consumption.

C++ CODE :CENTRAL LIMIT THEOREM

```
#include <iostream>

#include <cmath>

using namespace std;

// STEP 1: Normal CDF (P(Z <= z))

// This gives the probability from Z-table

double normalCDF(double z)

{

    return 0.5 * (1 + erf(z / sqrt(2.0)));

}

// STEP 2: Standard Error Formula

// SE = sigma / sqrt(n)

double getStandardError(double sigma, double n)

{

    return sigma / sqrt(n);

}

// STEP 3: Z-Score Formula

// Z = (value - mean) / SE

double getZScore(double value, double mean, double SE)

{

    return (value - mean) / SE;

}

// MAIN PROGRAM

int main()
```


{
double mu, sigma, n;
cout << "===== " << endl;
cout << " CLT SOLVER - Central Limit Theorem " << endl;
cout << "===== " << endl;
// --- Input ---
cout << "\nEnter Population Mean (mu) : ";
cin >> mu;
cout << "Enter Population Std Deviation (s) : ";
cin >> sigma;
cout << "Enter Sample Size (n) : ";
cin >> n;
// --- Calculate Standard Error ---
double SE = getStandardError(sigma, n);
cout << "\n===== " << endl;
cout << " RESULTS" << endl;
cout << "===== " << endl;
cout << "\n[CLT] Sampling Distribution:" << endl;
cout << " X_bar ~ N(" << mu << " , " << SE << ")" << endl;
cout << "\n[A] Standard Error (SE):" << endl;
cout << " SE = " << sigma << " / sqrt(" << n << ")" << endl;
cout << " SE = " << SE << endl;
// --- Choose what to calculate ---

int choice;
cout << "\n===== " << endl;
cout << "What do you want to calculate?" << endl;
cout << " 1) $P(\bar{X} > \text{value})$ " << endl;
cout << " 2) $P(\bar{X} < \text{value})$ " << endl;
cout << " 3) $P(a < \bar{X} < b)$ " << endl;
cout << " 4) All three" << endl;
cout << "Enter choice (1/2/3/4): ";
cin >> choice;
double value, a, b, Z, Z1, Z2, prob;
// --- $P(\bar{X} > \text{value})$ ---
if (choice == 1 choice == 4)
{
cout << "\nEnter value for $P(\bar{X} > ?)$: ";
cin >> value;
Z = getZScore(value, mu, SE);
prob = 1 - normalCDF(Z);
cout << "\n[B] $P(\bar{X} > $ " << value << "):" << endl;
cout << " Z = (" << value << " - " << mu << ") / " << SE << " = " << Z << endl;
cout << " $P(Z > $ " << Z << ") = 1 - $P(Z \leq $ " << Z << ")" << endl;
cout << " $P(Z > $ " << Z << ") = 1 - " << normalCDF(Z) << endl;
cout << " >>> Answer = " << prob << " (" << prob * 100 << "%)" << endl;
}

// --- $P(\bar{X} < \text{value})$ ---
if (choice == 2 choice == 4)
{
cout << "\nEnter value for $P(\bar{X} < ?)$: ";
cin >> value;
Z = getZScore(value, mu, SE);
prob = normalCDF(Z);
cout << "\n[C] $P(\bar{X} < " << \text{value} << ")$:" << endl;
cout << " Z = (" << value << " - " << mu << ") / " << SE << " = " << Z << endl;
cout << " $P(Z < " << Z << ")$ = " << prob << endl;
cout << " >>> Answer = " << prob << " (" << prob * 100 << "%)" << endl;
}
// --- $P(a < \bar{X} < b)$ ---
if (choice == 3 choice == 4)
{
cout << "\nEnter lower value (a): ";
cin >> a;
cout << "Enter upper value (b): ";
cin >> b;
Z1 = getZScore(a, mu, SE);
Z2 = getZScore(b, mu, SE);
prob = normalCDF(Z2) - normalCDF(Z1);
cout << "\n[D] $P(" << a << " < \bar{X} < " << b << ")$:" << endl;
cout << " Z1 = (" << a << " - " << mu << ") / " << SE << " = " << Z1 << endl;

cout << " Z2 = (" << b << " - " << mu << ") / " << SE << " = " << Z2 << endl;
cout << " P(Z1 < Z < Z2) = P(Z < " << Z2 << ") - P(Z < " << Z1 << ")" << endl;
cout << " = " << normalCDF(Z2) << " - " << normalCDF(Z1) << endl;
cout << " >>> Answer = " << prob << " (" << prob * 100 << "%)" << endl;
}
cout << "\n===== " << endl;
cout << " DONE!" << endl;
cout << "===== " << endl;
return 0;
}

OUTPUT

=====

CLT SOLVER - Central Limit Theorem

=====

Enter Population Mean (mu) : 18

Enter Population Std Deviation (s) : 6

Enter Sample Size (n) : 49

=====

RESULTS

=====

[CLT] Sampling Distribution:

$\bar{X} \sim N(18, 0.857143)$

[A] Standard Error (SE):

$SE = 6 / \sqrt{49}$

$SE = 0.857143$

=====

What do you want to calculate?

1) $P(\bar{X} > \text{value})$

2) $P(\bar{X} < \text{value})$

3) $P(a < \bar{X} < b)$

4) All three

Enter choice (1/2/3/4): 4

Enter value for $P(\bar{X} > ?)$: 20

[B] $P(\bar{X} > 20)$:

$$Z = (20 - 18) / 0.857143 = 2.33333$$

$$P(Z > 2.33333) = 1 - P(Z \leq 2.33333)$$

$$P(Z > 2.33333) = 1 - 0.990185$$

$$>>> \text{Answer} = 0.00981533 \text{ (0.981533\%)}$$

Enter value for $P(\bar{X} < ?)$: 40

[C] $P(\bar{X} < 40)$:

$$Z = (40 - 18) / 0.857143 = 25.6667$$

$$P(Z < 25.6667) = 1$$

$$>>> \text{Answer} = 1 \text{ (100\%)}$$

Enter lower value (a): 16

Enter upper value (b): 20

[D] $P(16 < \bar{X} < 20)$:

$$Z1 = (16 - 18) / 0.857143 = -2.33333$$

$$Z2 = (20 - 18) / 0.857143 = 2.33333$$

$$P(Z1 < Z < Z2) = P(Z < 2.33333) - P(Z < -2.33333)$$

$$= 0.990185 - 0.00981533$$

$$>>> \text{Answer} = 0.980369 \text{ (98.0369\%)}$$

=====

DONE!

=====